TÉCNICO
UNIVERSIDADE
DE LISBOA

0.2 · SHIP AND OCEAN SYSTEMS DESIGN

Ship Design Project

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Structure of the Project

Ten **common deliverables (D1–D10)**, extended by **two (2) design specializations** per group.

The project is group work done outside class. The in-class exercises are individual practice on a reference ship and are not graded.

COMMON TO THE WHOLE PROJECT

Baseline Project (A or B)

E.G. GROUP 1 · BASELINE A

Preliminary Stability + Damage Probabilistic Calculations

E.G. GROUP 2 · BASELINE B

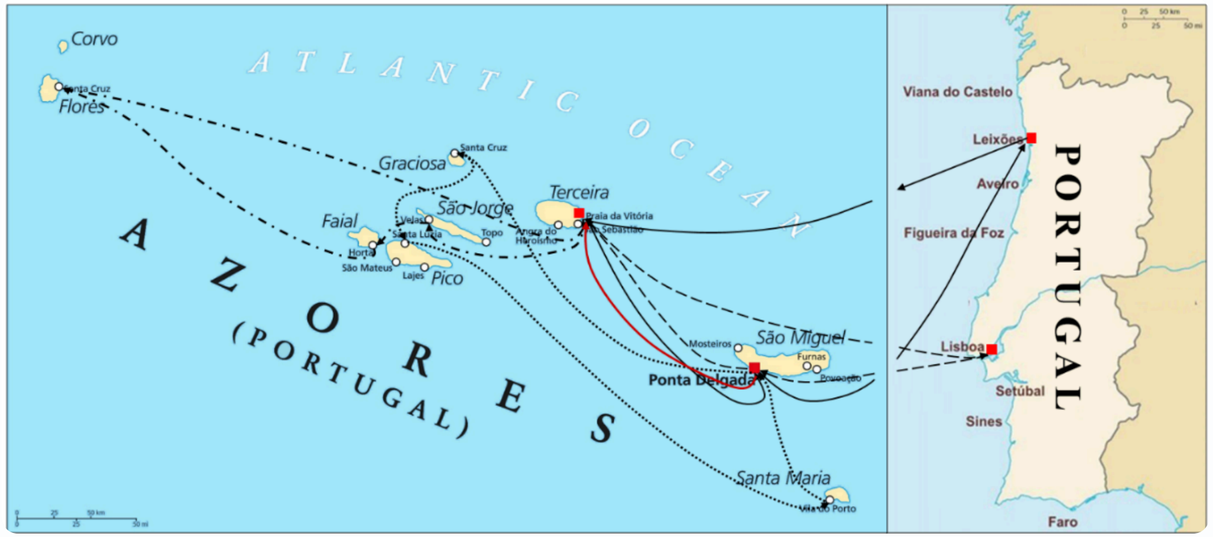
Electrical Load Balance + Alternative Fuel and Energy Storage

AVAILABLE DESIGN SPECIALIZATIONS

SP01 Longitudinal Strength Calculation	**SP02** Hull-Propeller Matching	**SP03** Electrical Load Balance	**SP04** Detailed Cost Estimation
SP05 Parametric Hull Shape Generation	**SP06** Main Engine Room & Steering Room Arrangement	**SP07** Preliminary Stability Calculation	**SP08** Hull Shape Optimization
SP09 Damage Probabilistic Calculations (Baseline A)	**SP10** Detailed Emissions Calculation	**SP11** Advanced Dimensional Analysis	**SP12** Alternative Fuel and Energy Storage

Two Ships for One Azores Container Service

Each group designs one of the two ships, assigned by the lecturer. Their exterior design should be similar.





Baseline A · Azores Express

1 300 TEU · mainland to islands, weekly · dual-fuel LNG



Baseline B · Azores Feeder

130 TEU · inter-island loops · battery-electric

Azores Express: Owner’s Requirements

Ship type
Container ship, forward superstructure, aft engine room, bow thruster, good seakeeping

Cargo
Containers · gearless

Target capacity
1 300 TEU

Route
Lisbon – Ponta Delgada – Praia da Vitória – Lisbon, one round voyage a week

Port restrictions
Probably none · to be checked by each group

Service speed
18–19 kn · 7–8 kn on the generators through a PTI

Fuel and propulsion
Dual-fuel HFO/LNG, single shaft · LNG in tanks or tank containers aft

Rules and regulations
Bureau Veritas · Portuguese flag · IMO conventions and codes

Design objective
Minimum propulsion power, with one round voyage per week

Environmental targets
EEDI · CII · ECA on the route

Specialization 1
Per group

Specialization 2
Per group

Alternative route: Leixões – Praia da Vitória – Ponta Delgada – Leixões, also one round voyage per week.

Azores Feeder: Owner’s Requirements

Ship type
Container ship on a large PSV-type hull (X-bow?), forward superstructure, bow thruster, good seakeeping

Cargo
Containers · geared, with its own container crane

Target capacity
130 TEU

Route
Two island loops plus one transit, all in one week (below)

Port restrictions
Probably none · to be checked by each group

Service speed
11–12 kn · 6–7 kn on the generator sets, to the mainland or in emergency

Fuel and propulsion
Swappable battery containers, aft on deck · two electric motors and shaft lines

Rules and regulations
Bureau Veritas · Portuguese flag · IMO conventions and codes

Design objective
Minimum battery capacity, with both loops in one week

Environmental targets
EEDI · CII · ECA only on the mainland transit

Specialization 1
Per group

Specialization 2
Per group

Loop 1: Ponta Delgada – Graciosa – Pico – Santa Maria – Ponta Delgada · **Loop 2:** Praia da Vitória – São Jorge – Faial – Flores – Praia da Vitória · plus the transit Ponta Delgada – Praia da Vitória. Batteries sized for one loop plus the transit.

Common Deliverables D1–D10

Deliver the work in a single ZIP file (Ex.: Group1_Project.zip), by e-mail, until 13 December 2026.

#	CONTENT	MUST DEMONSTRATE
D1	**Design Basis**	Owner's requirements · Port limits · Environmental targets
D2	**Reference Ship Database**	Similar ships under 5 years old · Sources · Regressions and bounds
D3	**Ship Model**	Inputs and outputs · Calculation sequence · Closed weight balance
D4	**Voyage and Financial Model**	Ship model outputs as input · Port restrictions · Revenue assumptions
D5	**Validation Note**	Reference ships · Error per quantity · Model corrected
D6	**Optimization Report**	Justified objective · Active constraints · Optimum vs requirements
D7	**Compliance Sheet**	EEDI and CII · Freeboard and stability · Draught at every port
D8	**General Arrangement and Hull Form**	A3 sketch: profile and midship section · Hydrostatics check
D9	**Specialization Documents (2)**	Calculation note · Required drawings · One per specialization
D10	**Datasheet and Presentation**	One-page A4 datasheet · Presentation of 15 minutes

> **Note:** What each deliverable must cover for your ship: slides 11–12 and your group's Project Scope Sheet, handed out in week 1. The class exercises are practice, not part of the project.

Specialization Content

#	DESIGN SPECIALIZATION	MUST DEMONSTRATE
SP01	**Longitudinal Strength Calculation**	Loads and bending moments · Section modulus check · Midship section drawing
SP02	**Hull-Propeller Matching**	Off-design operating points · Propeller selection · Stern arrangement drawing
SP03	**Electrical Load Balance**	Consumers per operating mode · Generator sizing · One-line diagram
SP04	**Detailed Cost Estimation**	Cost breakdown by group · Unit cost sources · Link to General Arrangement
SP05	**Parametric Hull Shape Generation**	Parametric hull definition · Coupling to the optimizer · Lines plan
SP06	**Engine Room & Steering Room Arrangement**	Machinery layout · Rudder and steering gear · Arrangement drawing
SP07	**Preliminary Stability Calculation**	Loading conditions · GZ curves against criteria · Tank and capacity plan
SP08	**Hull Shape Optimization**	Hull variables and objective · Resistance gain · Initial and optimized lines
SP09	**Damage Probabilistic Calculations**	Baseline A only · Watertight subdivision · A ≥ R · Subdivision plan
SP10	**Detailed Emissions Calculation**	Fuel use per mode · CO2, SOx, NOx per year · Machinery arrangement
SP11	**Advanced Dimensional Analysis**	Regression equations for dimensions and ratios · Fit quality and validity range
SP12	**Alternative Fuel and Energy Storage**	Energy per voyage · Tank or battery sizing · Rules check

Note: Each specialization has its own self-study deck. Your group's two are written on its Project Scope Sheet.

RESOURCES

Software

Calculations and programming

Python

High-level programming language

Google Colab or VS Code

Integrated development environment (IDE)

Excel

Spreadsheet and Solver, if Python is not used

3D modelling and drafting

Rhinoceros 3D

Surface modeling

Maxsurf Modeler

Hull form modeling

DELFTship (free)

Hull form modeling

AutoCAD

Computer Aided Design system

> **Note:** Bring a computer to every class: each student does the exercises individually, in Python or Excel.

The Reference Ship: Owner’s Brief

Every in-class exercise uses this ship: the 10,000 t cargo ship of Exercise 1.4. Your project uses your Baseline ship (slide 4 or 5).

Requirement	Value
Type	General cargo ship, single screw, diesel, machinery and deckhouse aft
Cargo	General cargo and bulk, free in and out (FIO)
Deadweight	10,000 t at the design draught
Service speed	14 kn (engine at 90% SMCR, 15% sea margin)
Route	Leixões – Funchal – Leixões, liner shuttle, 350 operating days a year
Cargo per round voyage	8,000 t southbound, 4,000 t northbound
Cargo handling	4,000 t/day at Leixões, 3,000 t/day at Funchal (shore cranes)
Hold capacity	At least 14,000 m³ (grain)
Draught limit	7.5 m at both ports
Fuel	VLSFO for the main engine, MGO for the generators
Rules	Class rules, SOLAS, MARPOL (EEDI for general cargo ships), Load Lines

EXERCISE 1.1 · INDIVIDUAL · 45 MIN · PRACTICE FOR D1

Turn the Reference Ship’s Brief into a Design Basis

1 Rewrite · 10 min

Put every line of the reference ship’s brief (slide 9) in a table: requirement, value, unit, source.

2 Tag · 10 min

Mark each line Given, Assumed or To be checked. Say who confirms the assumed ones.

3 Ports · 8 min

For Leixões and Funchal: which limits you need (draught, length, beam, berth) and where you will find them.

4 Gaps · 7 min

What does a first design need that the brief does not give? Crew, range, flag, stowage factor… Write them as questions.

Check: the worked answer is shown at the end (Exercise Answers – Week 1). Not graded.
Your project: your group does the same for its own ship outside class and completes D1 (see your Project Scope Sheet).

Azores Express: What Each Deliverable Must Cover

Deliverable	Scope for the Express
D1 Design Basis	1,300 TEU at 10 t per TEU on average (13,000 t cargo) · 18–19 kn · weekly round voyage · limits at Lisbon, Ponta Delgada, Praia da Vitória (and Leixões)
D2 Reference ships	At least 10 gearless container ships of about 900–1,800 TEU, recent where possible · Appendix A.3
D3 Ship Model	Volume-carrier sequence (Table 1.9) · dual-fuel machinery · LNG storage aft in the weights
D4 Voyage and finance	One round voyage a week: sea and port time, LNG and HFO use, fuel and port costs, freight revenue
D5 Validation	Model against the reference ships: L, B, T, Δ, lightship, power
D6 Optimization	Minimum propulsion power that keeps the weekly schedule
D7 Compliance	EEDI and CII as targets · ECA near the mainland (Tier III, 0.10% S) · freeboard · stability with deck stacks · draught at each port
D8 GA and hull form	Bays, rows and tiers · forward superstructure · LNG storage aft · midship section

Azores Feeder: What Each Deliverable Must Cover

Deliverable	Scope for the Feeder
D1 Design Basis	130 TEU at 10 t per TEU (1,300 t cargo) · battery containers extra, mass from supplier data · 11–12 kn · two loops plus transit in a week · limits at all eight ports
D2 Reference ships	At least 8 small geared container ships and battery ships · other families as the chosen hull needs
D3 Ship Model	Volume-carrier sequence · electric drive, two shafts · batteries as deadweight · crane in lightship
D4 Voyage and finance	Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs
D5 Validation	Model against the reference families the group chose, with the choice justified
D6 Optimization	Minimum battery capacity that completes both loops in one week
D7 Compliance	EEDI and CII: show why they do not apply, report energy per TEU-mile · battery rules (BV) · stability while the crane lifts · draught at each port
D8 GA and hull form	Deck container layout and crane reach · battery slots aft · hull form of the group's choice (a PSV-type hull is suggested)